

RFM Customer Segmentation Analysis

MySQL and Python project | 287 customers

Business question

Which customers drive revenue, which are at risk of leaving, and where should retention effort go first?

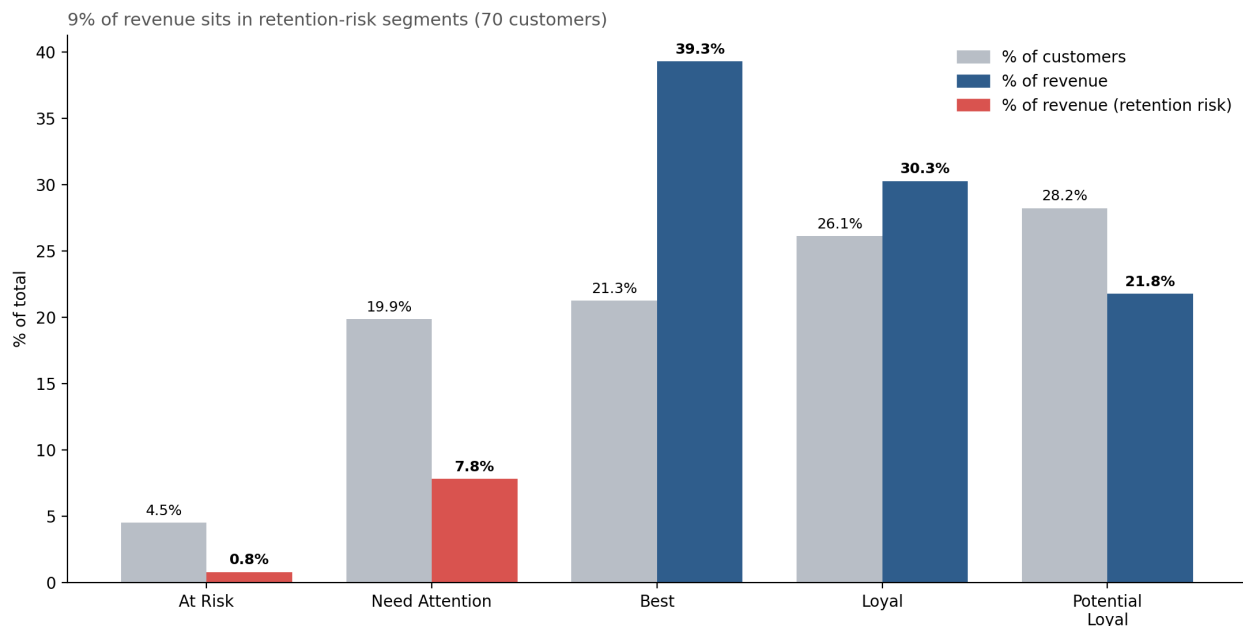
Method

Scored each customer on Recency, Frequency and Monetary value and grouped them into five segments: Best, Loyal, Potential Loyal, Need Attention and At Risk. Used MySQL (window functions, temporary tables) to measure each segment's size, revenue share and management priority, and Python (pandas, matplotlib) to visualize the results.

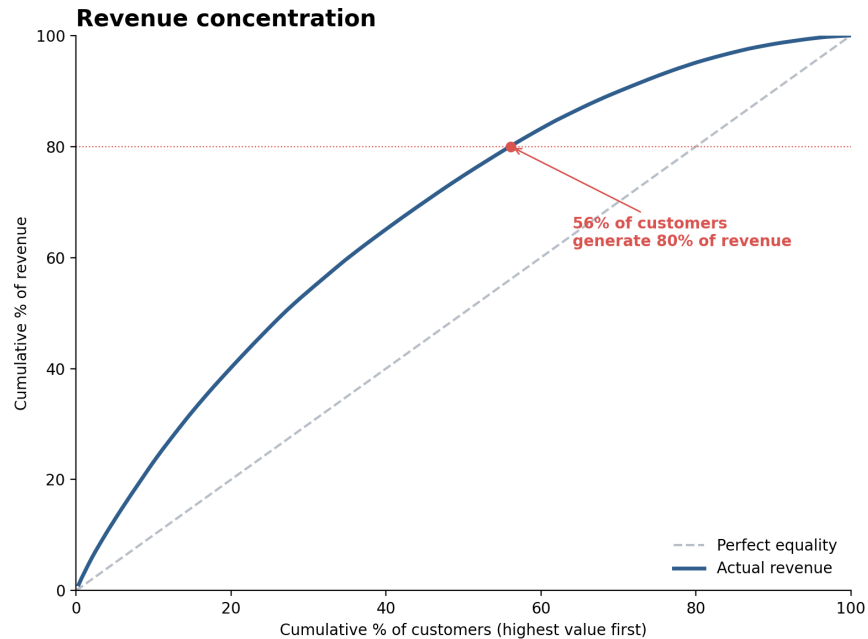
Key findings

- **1. A minority of customers drive most of the revenue.** Best Customers are 21.3% of customers but generate 39.3% of revenue. Together with Loyal Customers they account for 69.6% of revenue from 47.4% of customers.
- **2. Revenue at risk is modest but worth acting on.** At Risk and Need Attention hold 8.6% of revenue across 70 customers (24.4% of the base). Most of that sits in Need Attention (7.8%). At Risk customers are 4.5% of the base but only 0.8% of revenue, so win-back effort should target the highest-value ones first.
- **3. Concentration is moderate, with a clear growth opportunity.** 56% of customers generate 80% of revenue. Potential Loyal is the largest segment (28.2% of customers) but contributes only 21.8% of revenue, making it the main upgrade target.

Best Customers: 21% of customers, 39% of revenue



Revenue concentration



Recommendations

- **Protect Best Customers:** they carry 39% of revenue, so losing them costs the most. Use loyalty perks and early access.
- **Prevent churn in Need Attention:** the largest exposed pool (7.8% of revenue). Run a preventive retention campaign before they lapse.
- **Win back selectively:** focus At Risk outreach on the top-value customers rather than the whole segment.
- **Develop Potential Loyal:** targeted offers to lift their frequency and spend toward Loyal levels.

Tools: MySQL, SQL window functions, Python, pandas, matplotlib